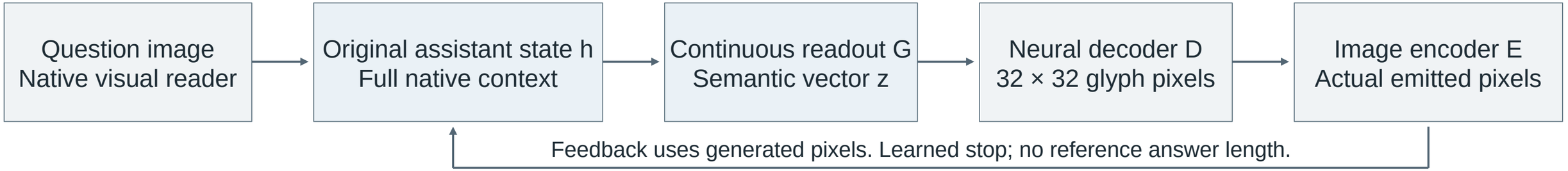
A complete-glyph semantic loop separates writing from continuation

a Evaluated language runtime: one whole glyph per event



b Completed primitive training

Offline source anchors + canonical glyph images
Primitive: train E/D; freeze readout/stopping.
Later: fit readout/stopping on 508 question-image histories.
4,096 anchored characters → 4,087 distinct image forms.

Direct: 4,048/4,087 exact; F1 0.999940
Roundtrip: 3,697/4,087 exact; F1 0.999634
Semantic cosine: 0.77735 on first 128; alignment remains weaker.
80.22 s / 0.728 GiB for the primitive only.

c Gold / direct decode / actual-image roundtrip



64 evenly spaced indices; all physical pixels retained. This is reconstruction, not answer accuracy.
Broader Chinese assessment: 1/16 exact, 11 caps. No runtime tokenizer, source text head, font renderer or glyph lookup.